CSE 225 Data Structures
Week #8

B
Quiz #3

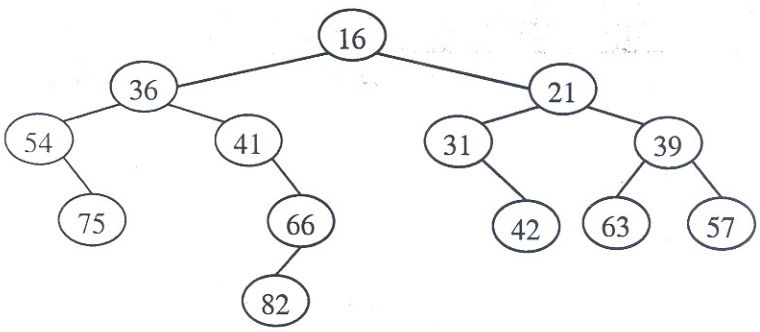
Borahan Tümer, Ph.D.
Fall 2006

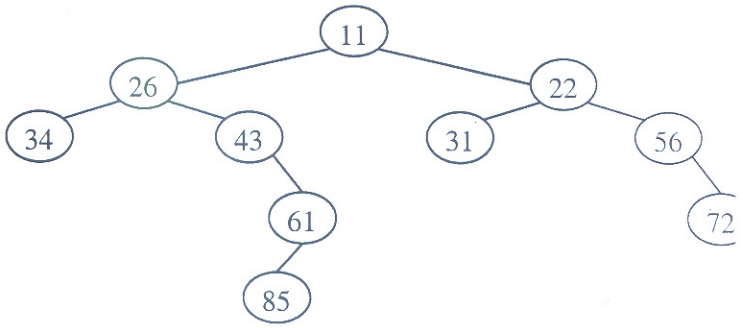
Q1.

Two heaps are given below.

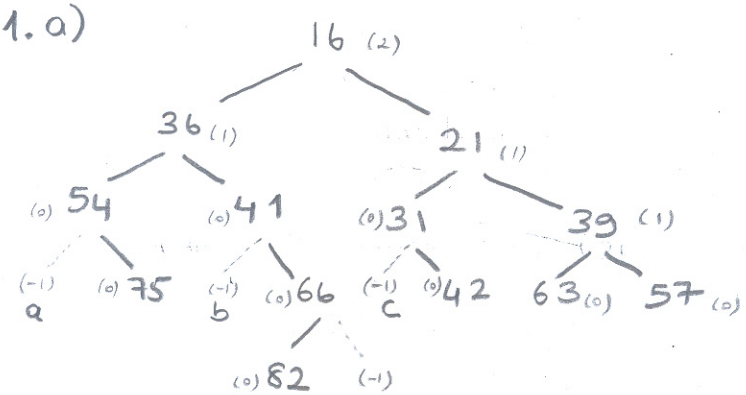
a) Are they leftist heaps?

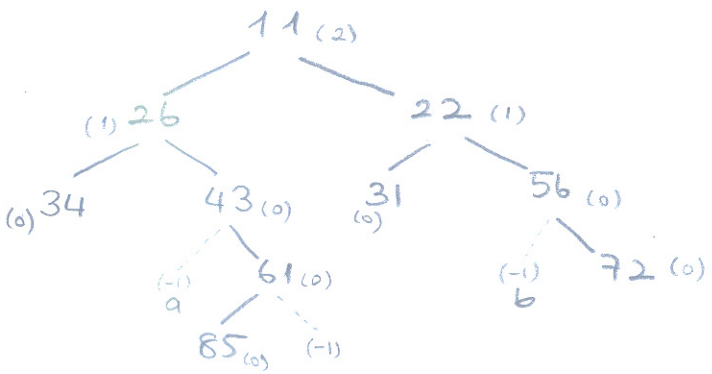
b) Merge them ***recursively*** in ***O(lg n)*** time where ***n*** is the number of elements in both heaps! Make any (or both if necessary) in case they are not LHs!





1. a)

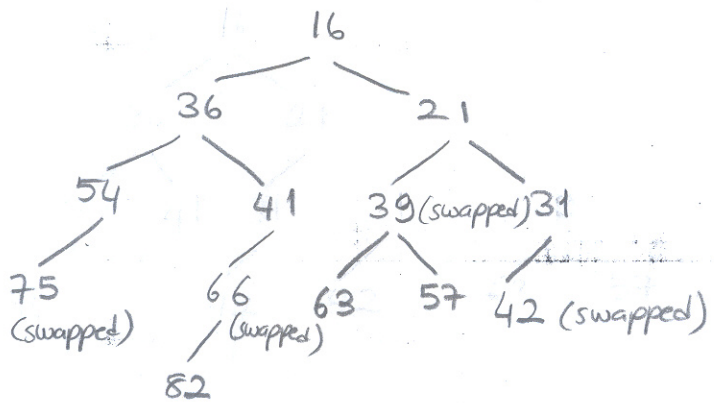


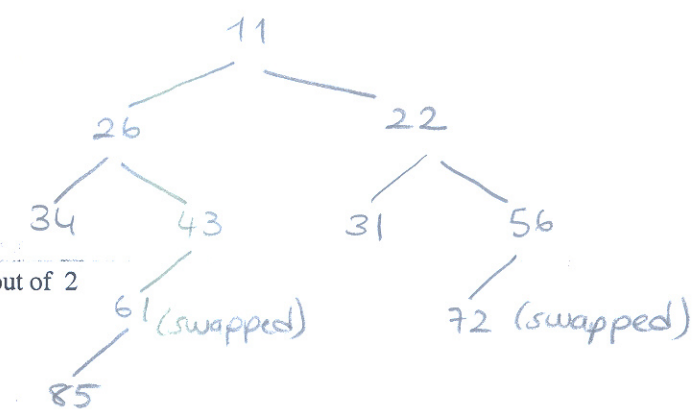


Not leftist: 1) Npl (31) < Npl (39)
2) Npl (a) < Npl (75)
3) Npl (b) < Npl (66)
4) Npl (c) < Npl (42)

Not leftist: 1) Npl (a) < Npl (61)
2) Npl (b) < Npl (72)

1. b) First, make the heaps leftist

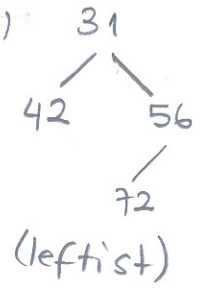




1.6) cont Then, merge the heaps. (8)

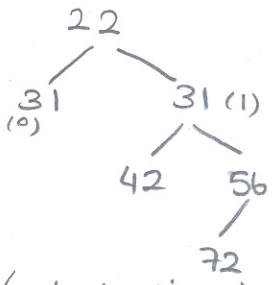
Comparison table: 16 - 11 (5)
16 - 22 (4)
21 - 22 (3)
31 - 22 (2)
31 - 56 (1)

1)



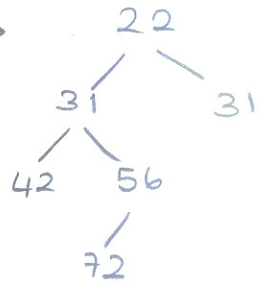
(leftist)

2)

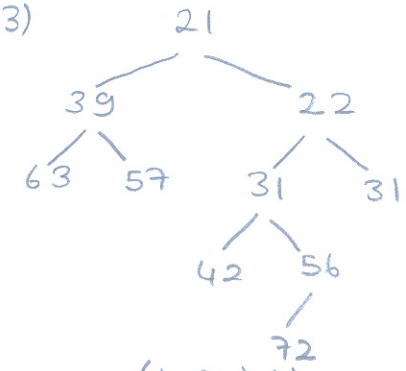


(not leftist)

→

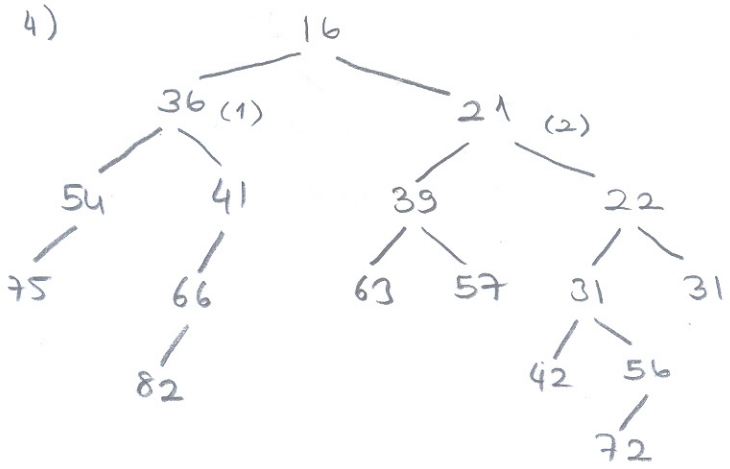


3)



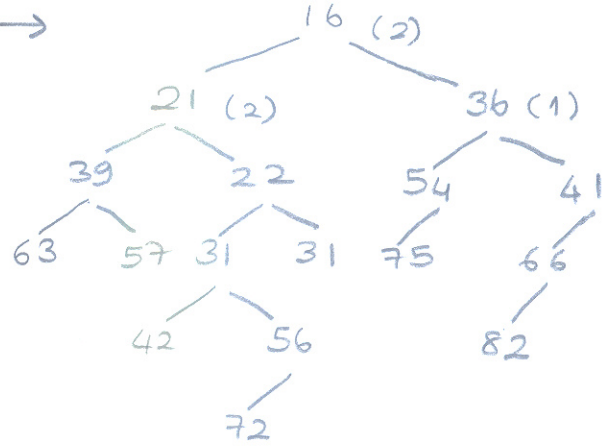
(leftist)

4)



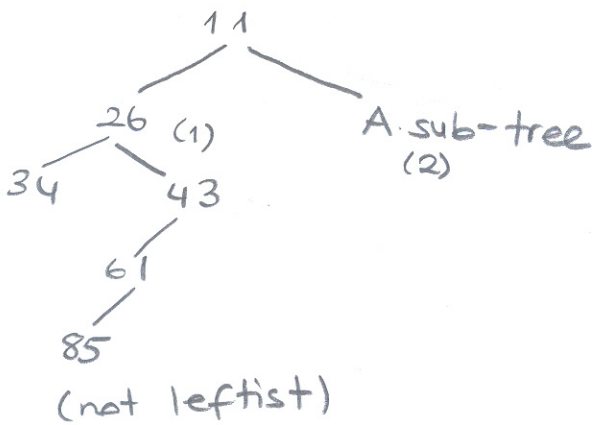
(not leftist)

→



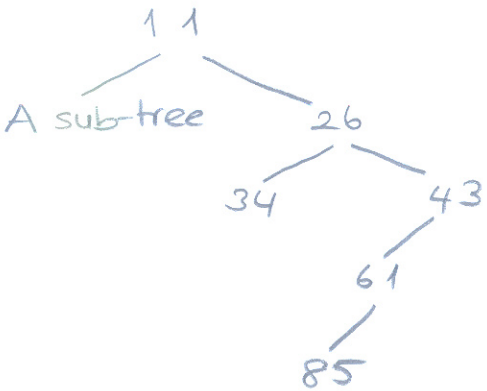
(A sub-tree)

5)



(not leftist)

→



(result)

Q2.

Consider a binomial heap of 1279 nodes.

a) How many binomial trees does it have?
b) What are these trees?
c) How many nodes does the "highest" (or "deepest") binomial tree have in its ***most crowded depth level***? (most crowded: en kalabalık)

2.a) $1279 = 2^{10} + 2^7 + 2^6 + 2^5 + 2^4 + 2^3 + 2^2 + 2^1 + 2^0$

So, it has 9 binomial trees

2.b) $B_0, B_1, B_2, B_3, B_4, B_5, B_6, B_7, B_{10}$

2.c.) • highest binomial tree is B_{10}.

• most crowded depth level is the one(s) in the middle of the tree, that is 5th level.

• it has:

$$\binom{k}{d} = \binom{10}{5} = \frac{10!}{5!\,5!} = 252 \text{ nodes}$$